

**LESSON 5-7 • SESSION 1 • GUIDED NOTES****Surface Area of Rectangular Prisms**

Learning target – I can find the surface area of a rectangular prism by adding the areas of its six faces. • 6.G.A.4

**WORDS TO KNOW***fill in the missing words*

| WORD                                           | WHAT IT MEANS                               | EXAMPLE              |
|------------------------------------------------|---------------------------------------------|----------------------|
| <b>surface area</b><br><i>área superficial</i> | The total ____ of all the faces of a solid. | add every face       |
| <b>congruent</b><br><i>congruente</i>          | Exactly the same ____ and shape.            | opposite faces match |
| <b>net</b><br><i>plantilla</i>                 | A flat ____ that folds into a solid.        | shows all 6 faces    |

**KEY IDEA***complete each sentence*

- A rectangular prism has \_\_\_\_ pairs of matching faces.
- $SA = 2lw + 2lh + \underline{\hspace{1cm}}$ .
- Surface area is measured in \_\_\_\_ units.

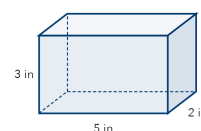
**WORKED EXAMPLE***fill in the blanks as we solve*

A closed box is 6 cm long, 4 cm wide, and 3 cm tall. Find its surface area.

**Step 1** Top:  $6 \times 4 = \underline{\hspace{1cm}}$  cm<sup>2</sup>. Front:  $6 \times 3 = 18$  cm<sup>2</sup>. End:  $4 \times 3 = \underline{\hspace{1cm}}$  cm<sup>2</sup>.**Step 2** Double each:  $48 + 36 + 24$ .**Answer**  $SA = \underline{\hspace{1cm}}$  cm<sup>2</sup>**YOU TRY***use the example as a model*

- Find the surface area of the prism.

*Show your work.*

**Answer** \_\_\_\_\_ in<sup>2</sup>

- Find the surface area of each cube.

**a.** edges 2 m \_\_\_\_\_ m<sup>2</sup>**b.** edges 10 in \_\_\_\_\_ in<sup>2</sup>**SUM IT UP***in your own words*

Surface area is different from volume because \_\_\_\_.

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# LESSON 5-7 • SESSION 2 • GUIDED NOTES

## Surface Area of Triangular Prisms

Learning target – I can find the surface area of a triangular prism using its net. · 6.G.A.4

### WORDS TO KNOW

fill in the missing words

| WORD                                                | WHAT IT MEANS                                        | EXAMPLE          |
|-----------------------------------------------------|------------------------------------------------------|------------------|
| <b>triangular prism</b><br><i>prisma triangular</i> | A solid with 2 triangle _____ and 3 rectangle faces. | a tent shape     |
| <b>lateral face</b><br><i>cara lateral</i>          | A side face that is _____ a base.                    | the 3 rectangles |
| <b>surface area</b><br><i>área superficial</i>      | The total area of all the _____.                     | add all 5 faces  |

### KEY IDEA

complete each sentence

- Add the areas of all \_\_\_\_\_ faces: 2 triangles and 3 rectangles.
- Each triangle uses  $\frac{1}{2} \times \text{base} \times \text{height}$ .
- Each rectangle is one triangle \_\_\_\_\_  $\times$  the prism's length.

### WORKED EXAMPLE

fill in the blanks as we solve

A prism is 6 ft long. Each triangle end has a base of 8 ft, a height of 3 ft, and two 5 ft sides. Find its surface area.

**Step 1** One end:  $\frac{1}{2} \times 8 \times 3 = \underline{\hspace{1cm}}$  ft<sup>2</sup>. Both ends: 24 ft<sup>2</sup>.

**Step 2** Rectangles:  $8 \times 6 + 5 \times 6 + 5 \times 6 = \underline{\hspace{1cm}}$  ft<sup>2</sup>.

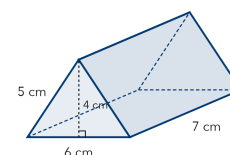
**Answer** **SA = 24 + 108 = \_\_\_\_\_ ft<sup>2</sup>**

### YOU TRY

use the example as a model

- Find the surface area of the triangular prism.

Show your work.



**Answer** \_\_\_\_\_ cm<sup>2</sup>

- Find each area.

**a.** 2 triangles, base 10 in, height 2  
in \_\_\_\_\_ in<sup>2</sup>

**b.** rectangle 4 in by 9 in \_\_\_\_\_ in<sup>2</sup>

### SUM IT UP

in your own words

For a triangular prism, I use the triangle's height for \_\_\_\_\_ and its sides for \_\_\_\_\_.

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